

PROJECT DESIGN PHASE : PROPOSED SOLUTION DOCUMENT

SZ METRO TICKET GENERATING SYSTEM IN SERVICENOW

Date: 31 July 2026Developer ID: 23JR1A05E4 (Individual)Developer Name: Shaik Zuveriya TabassumMax Marks: 2 Marks

END-TO-END SYSTEM SOLUTION ARCHITECTURE FLOW

S.NO.	PARAMETER	DESCRIPTION
1.	Problem Statement (Problem to be solved)	Heavy station congestion, long ticket counter queues during peak commuting hours, manual payment delays, and physical paper/token loss lead to missed metro trains and poor passenger experiences in urban transit systems.
2.	Idea / Solution Description	The SZ Metro Ticket Generating System is a cloud-native ServiceNow solution featuring: • Service Portal Self-Service: Passengers book single/return tickets via mobile web portal in under 30 seconds without installing an app. • Dynamic Fare Matrix Engine: Script Includes (GlideAjax) automatically calculate exact fares based on source and destination station distance tables. • Encrypted QR Ticket & Workflows: Generates dynamic, time-sensitive scannable QR payloads with automated lifecycle triggers (Active → Used/Expired). • Inspector Scan Interface: Mobile portal for station staff to quickly verify ticket validity at turnstiles.
3.	Novelty / Uniqueness	• Zero Native App Installation: Accessible directly via lightweight browser-based ServiceNow Service Portal. • Native Platform Orchestration: Built fully on enterprise PaaS (ServiceNow), leveraging unified workflows, automated notification gateways (Email/SMS), and strict Role-Based Access Control (ACLs). • Real-time Status Automation: Automated Business Rules dynamically invalidate used or expired QR payloads to prevent duplicate entries or fare fraud.
4.	Social Impact / Customer Satisfaction	• Zero Queue Waiting Time: Eliminates crowded station queues, drastically saving daily commute time. • Eco-Friendly Paperless Travel: Replaces plastic tokens and paper tickets with digital QR passes. • Enhanced User Experience: Delivers high commuter satisfaction through transparent fare previews and instant automated receipts.
5.	Business Model (Revenue Model)	• Convenience Fee Micro-Transactions: Small digital service fee (e.g., ₹1-2 per online ticket) retained per booking. • Platform Licensing for Transit Agencies: Enterprise PaaS solution leased to municipal metro rail authorities on a subscription model. • Digital Partner Advertising: Non-intrusive promotional ads displayed on digital QR receipt pages.
6.	Scalability of the Solution	• Elastic Cloud Infrastructure: Hosted on ServiceNow Enterprise Cloud, capable of seamlessly processing thousands of concurrent ticket requests during peak morning rush hours. • Multi-City Transit Expansion: Easily configurable data tables allow adding new metro lines, routes, and fare matrices without code refactoring.

Document Version: 1.0.0 | Project: SZ Metro Ticket Generating System in ServiceNow | Developer: Shaik Zuveriya Tabassum (23JR1A05E4)